

A small pupil with ptosis points to sympathetic loss

Horner syndrome produces ipsilateral miosis, mild ptosis and dilation lag.

The sympathetic pathway

First order

Brainstem to C8-T2; lateral medullary or spinal disease.

Second order

Lung apex to cervical ganglion; Pancoast or neck lesions.

Third order

Along the ICA; carotid dissection or cavernous disease.

Clinical meaning

Horner links ipsilateral mild ptosis and miosis with dilation lag. Anisocoria increases in the dark because the smaller pupil fails to dilate. First- or second-order lesions may affect sweating; postganglionic disease may not. Associated pain, pulmonary history or neurological signs help trace the cause. Isolation does not establish a benign diagnosis.

A painful new Horner pattern is an emergency

The timing and pain matter as much as the degree of pupil asymmetry.

Identify the abnormal pupil

Compare bright light with darkness

Greater difference in darkness

The smaller pupil has impaired dilation, as in Horner.

Greater difference in bright light

The larger pupil has impaired constriction.

Clinical meaning

New painful Horner warrants urgent neurological and vascular assessment for carotid dissection. Adie has a large tonic pupil with light-near dissociation; Holmes-Adie adds reduced tendon reflexes. Argyll Robertson pupils are small and bilateral. A large pupil with ptosis and ophthalmoplegia requires third nerve assessment rather than an Adie label.